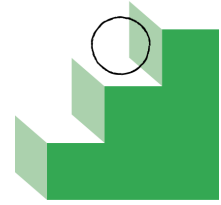


User Needs + Defining Success

Chapter worksheet



Instructions

Block out time to get as many cross-functional leads as possible together in a room to work through these exercises & checklists.

Exercises

1. Evidence of user need [multiple sessions]

Gather existing research and make a case for using AI to solve your user need.

2. Augmentation versus automation [multiple sessions]

Conduct user research to understand attitudes around automation versus augmentation.

3. Design your reward function [~1 hour]

Weigh the trade offs between precision and recall for the user experience.

4. Define success criteria [~1 hour]

Agree on how to measure if your feature is working or not, and consider the second order effects.

1. Evidence of user need

Before diving into whether or not to use AI, your team should gather user research detailing the problem you're trying to solve. The person in charge of user research should aggregate existing evidence for the team to reference in the subsequent exercises.

User research summary

List out the existing evidence you have supporting your user need. Add more rows as needed.

Date	Source	Summary of findings
2025-01-15	Interviews with financial analysts	Analysts spend 20–40 hours manually researching companies (Wikipedia, SEC filings, news) before making early-stage investment decisions.
2025-01-20	Internal workflow audit	Repetitive, high-effort data gathering is not standardized, making due diligence inconsistent across teams.
2025-01-25	Desk research on startup VCs	80% of analysts rely on fragmented sources (Google, PDFs, spreadsheets), which increases the risk of missed signals or outdated information.
2025-01-28	Product discovery interviews	All respondents expressed interest in a tool that consolidates filings and market signals automatically but want control over what's included.
2025-02-02	Literature & industry reports	Generative AI and RAG-based due diligence tools have shown time savings of 50–70%, but trust and explainability remain critical.

Make a case for and against your AI feature

Meet as a team, look at the existing user research and evidence you have, and detail the user need you're trying to solve.

Next, write down a clear, focused statement of the user need and read through each of the statements below to identify if your user need is a potential good fit for an AI solution.

At the end of this exercise your team should be aligned on whether AI is a solution worth pursuing and why.

How might we solve _____{ our user need }_____?

Can AI solve this problem in a unique way?

Our user need: Analysts need a faster and more reliable way to collect, validate, and synthesize company intelligence from multiple sources so that they can make better investment decisions without spending hours on repetitive manual research.

AI probably better	AI probably not better
☒ The core experience requires recommending different content to different users. ☒ The core experience requires prediction of future events. ☒ Personalization will improve the user experience. ☒ User experience requires natural language interactions. ☒ Need to recognize a general class of things that is too large to articulate every case. ☒ Need to detect low occurrence events that are constantly evolving. ☒ An agent or bot experience for a particular domain. ☒ The user experience doesn't rely on predictability.	☐ The most valuable part of the core experience is its predictability regardless of context or additional user input. ☐ The cost of errors is very high and outweighs the benefits of a small increase in success rate. ☐ Users, customers, or developers need to understand exactly everything that happens in the code. ☐ Speed of development and getting to market first is more important than anything else, including the value using AI would provide. ☐ People explicitly tell you they don't want a task automated or augmented.



We think AI { **can / can not** } help solve _____{ **user need** }_____, because

Yes, by using AI retrieval and summarization models, we can automatically gather and validate research from multiple sources, reducing manual effort, improving information consistency, and providing explainable outputs that analysts can trust and verify.

2. Augmentation versus automation

Conduct research to understand user attitudes

If your team has a hypothesis for why AI is a good fit for your user's need, conduct user research to further validate if AI is a good solution through the lens of automation or augmentation.

If your team is light on field research for the problem space you're working in, contextual inquiries can be a great method to understand opportunities for automation or augmentation.

Below are some example questions you can ask to learn about how your users think about automation and augmentation.

Research protocol questions

- If you were helping to train a new coworker for a similar role, what would be the most important tasks you would teach them first?

Collecting company data from structured (e.g., SEC) and unstructured (e.g., news, Wikipedia) sources, validating source credibility, summarizing filings, and preparing a quick snapshot before deep analysis.

- Tell me more about that action you just took, is that an action you repeat:

- ☒ Hourly
- ☐ Daily
- ☐ Weekly
- ☐ Monthly
- ☐ Quarterly
- ☐ Annually

- If you had a human assistant to work with on this task, what, if any, duties would you give them to carry out?

Delegating data gathering, cleaning, tagging relevant sections of filings, extracting key financial metrics, and compiling summaries for review.



If going to meet your users in context isn't feasible, you can also look into prototyping a selection of automation and augmentation solutions to understand initial user reactions.

The [Triptech method](#) is an early concept evaluation method that can be used to outline user requirements based on likes, dislikes, expectations, and concerns.

Research protocol questions

- Describe your first impression of this feature.

It feels like an intelligent research assistant that could save me hours collecting and cleaning data, as long as I can verify the sources. I like the idea of having summarized filings and news in one place.

- How often do you encounter the following problem: [insert problem/need statement here]?
 - Daily
 - ⊗ Often (a few times a week)
 - ⊗ Sometimes (a few times a month)
 - Rarely (a few times a year)
 - Never
- How important is it to address this need or problem?
 - Not at all important
 - Somewhat important
 - Moderately important
 - ⊗ Very important
 - ⊗ Extremely important

3. Design your reward function

Once your team has had a chance to digest your recent research on user attitudes towards automation and augmentation, meet as a team to design your AI's **reward function**. You'll revisit this exercise as you continue to iterate on your feature and uncover new insights about how your AI performs.

Use the template below to list out instances of each reward function dimension.

Reward function template

		Prediction	
		Positive	Negative
Reference	Positive	True Positive {Example 1} {Example 2} {Example 3}	False Negative {Example 1} {Example 2} {Example 3}
	Negative	False Positive {Example 1} {Example 2} {Example 3}	True Negative {Example 1} {Example 2} {Example 3}

Take a look at the false positives and false negatives your team has identified.

- If your feature offers the most user benefit for **fewer false positives**, consider optimizing for **precision**.
- If your feature offers the most user benefit for **fewer false negatives**, consider optimizing for **recall**.

Our AI model will be optimized for ^{precision}_____ { **precision / recall** } _____
because _____ { **user benefit** } _____

Analysts gain the most value when retrieved insights are reliable, relevant, and trustworthy, even if that means missing a few secondary items. False positives can erode confidence and increase manual verification time.

We understand that the tradeoff for choosing this method means our
model will _____ { **user impact** } _____

_____ occasionally miss some less-critical insights or emerging patterns (false negatives).
However, this ensures that every surfaced insight has high credibility and contextual
value, maintaining user trust and reducing cognitive load for analysts. _____

4. Define success criteria

Now that you've done the work to understand whether AI is a good fit for your user need and identified the tradeoffs of your AI's reward function, it's time to meet as a team to define success criteria for your feature. Your team may come up with multiple metrics for success by the end of this exercise.

By the end of this exercise, everyone on the team should feel aligned on what success looks like for your feature, and how to alert the team if there is evidence that your feature is failing to meet the success criteria.

Success metrics framework

Start with this template and try a few different versions:

If __{ **specific success metric** }__
for __ { **your team's specific AI driven feature** }__
{ **drops below/goes above** }__ { **meaningful threshold** }__
we will __{ **take a specific action** }__.

Version 1

If precision of retrieved research insights for the Automated Due Diligence Agent drops below 85%, we will review the retrieval and filtering logic, adjust vector similarity thresholds, and retrain the model with more curated examples.

Version 2

If average analyst research time per company profile for the Due Diligence workflow does not decrease by at least 40% compared to the manual baseline, we will analyze friction points in the pipeline and refine automation for data collection and summarization steps.

Version 3

If user trust score in AI-generated insights (measured via feedback surveys) for the Market Intelligence Agent falls below 4.0/5, we will conduct user interviews, improve explainability (source transparency, summary confidence), and retrain with cleaner validation data.

Statement iteration

Take each version through this checklist:

- ☒ Is this metric meaningful for all of our users?
 - ☒ How might this metric negatively impact some of our users?
- ☒ Is this what success means for our feature on day 1?
 - ☒ What about day 1,000?

Final version

If the precision of AI-curated research insights for the Automated Due Diligence Agent drops below 85% or user trust rating falls below 4.0/5, we will pause automated updates, run an error analysis on retrieval and summarization outputs, and refine both the validation schema and dataset sources.

Schedule regular reviews

Once you've agreed upon your success metric(s), put time on the calendar to hold your team accountable to regularly evaluate whether your feature is progressing towards and meeting your defined criteria.

Success metric review

Date: Every 2 months post-deployment

Attendees: Project Team Members — Hariharan Chandrasekar, Saumith Devarasetty, Niranjan Sathish, Harshit Kumar Taneja, Akshata Krishnamohan, Thanya Mysore Santhosh